#1 · 12.7% of marks

Ch 9 — Derivatives Analysis & Valuation

Tested every attempt: strangle / straddle payoffs, index-futures beta hedge, Black-Scholes, real options (risk-neutral), futures + options combos. Theory: real vs financial options.

① Users & Forward vs Futures (SM)

Users: *corporations — hedge currency & inventory risk · individual investors — speculation, hedging, yield enhancement · institutional investors — hedging asset allocation, yield enhancement, arbitrage · dealers — hedging, position taking, exploiting inefficiencies, dealer spreads.*

Feature	Forward	Futures
Trading	Personal basis / telephone	Competitive arena
Size	Individually tailored	Standardised
Exchange	OTC	Organised exchange
Settlement	On agreed date	Daily via clearing house
Delivery date	As agreed; actual delivery	Cyclical; delivery rarely takes place
Transaction cost	Bid-ask spread	Brokerage on buy & sell
Marking to market	No	Yes — daily in margin account
Margins	Not required	Required
Credit risk	Borne by each party	Not a concern (clearing house)

Basis = Spot − Futures. *Normal (**contango**) market: futures > spot (full cost of carry) → basis negative. **Backwardation / inverted**: spot > futures (factors other than carry) → basis positive. Basis → 0 at expiry = **convergence**.*

② Futures pricing — cost of carry

Situation	Formula
Basic (discrete)	$F = S + \text{cost of carry} - \text{dividend} = S(1+r)^t - D$
Continuous compounding	$F = S \cdot e^{rt}$
Known dividend	$F = (S - \text{PV of dividend}) \cdot e^{rt}$
Index with dividend yield y	$F = S \cdot e^{(r-y)t}$
Commodity with storage cost	$F = (S + \text{PV storage}) \cdot e^{rt}$ (convenience yield reduces F)

Actual F > Theoretical F

*Futures overpriced → **cash-and-carry**: borrow, **buy spot, sell futures**; deliver at expiry.*

Actual F < Theoretical F

*Futures underpriced → **reverse cash-and-carry**: **short-sell spot, invest, buy futures**.*

*SM: Future price = Spot + carrying cost − returns (dividends etc.). RTP Jan 2026: S = 125, r 8%, D = 4 → F = 125 + 10 − 4 = **₹131**; contract (1,000 shares) ₹1,31,000. Price falls 6% → S = 117.50 → F = 122.90 → long loses ₹8,100.*

REPEAT R17

③ Margins & mark-to-market

- → **Initial margin = μ + 3σ** of daily changes in contract value (SM illustration: μ ₹10,000 + 3 × ₹2,000 = ₹16,000; maintenance 75% = ₹12,000)
- → **Maintenance margin** (usually lower); balance falls below it → **margin call** to restore to **initial** margin
- → Daily MTM: long gains when price rises; losses debited to margin account. Cumulative MTM = (final − entry price) × lot

④ Beta hedge — formulas

No. of contracts to fully hedge = (Portfolio value × β_p) ÷ (Futures price × lot size)

Change beta from β_1 to β_2: contracts = ($\beta_2 - \beta_1$) × Portfolio value ÷ (Futures price × lot) — negative = **sell**, positive = **buy**

Long stock + short stock: hedge each leg separately (sell futures for long leg, buy for short leg) or net the beta-weighted values

Example	Working
Portfolio ₹1 cr, β 1.2; Nifty futures 22,000; lot 75	1,00,00,000 × 1.2 ÷ (22,000 × 75) = 7.27 → **sell 7 contracts**
Reduce β 1.2 → 0.6 (same data)	(0.6 − 1.2) × 1 cr ÷ 16,50,000 = −3.64 → sell **4**
RTP Jan 2026: portfolio 9,94,450, 5 Nifty contracts sold at 8,767.07 × 25 for full hedge	Implied β = 5 × 8,767.07 × 25 ÷ 9,94,450 = **1.102**; 1.102 = 0.9 × 1.10 + 0.1 × β_{10} → β_{10} = **1.12**

⚠ TRAP

Use the **futures** price (not spot index) in the denominator. Lot size is whatever the question gives — don't assume. Profit on hedge = (sell price − buy-back price) × lot × contracts; overall P/L = cash position change + futures P/L.

⑤ Options vs single stock futures (SM 6.7)

Point	Options	Stock futures
Nature	Buyer has right, not obligation	Long must square off at / before expiry
Movement of market	Premium moves with **delta**; may move independently of stock	Follows the underlying much more closely
Price of investing	Premium = **sunk cost**; lost if not ITM at expiry	Initial margin + maintenance (MTM) margin

7 Basics

	Call (right to BUY at K)	Put (right to SELL at K)
In the money	$S > K$	$S < K$
At the money	$S = K$	$S = K$
Out of the money	$S < K$	$S > K$
Intrinsic value	$\max(S - K, 0)$	$\max(K - S, 0)$
Time value	Premium − intrinsic value (all premium is time value if OTM / ATM)	
Buyer's net pay-off	$\max(S_T - K, 0)$ − premium	$\max(K - S_T, 0)$ − premium
Writer's pay-off	Exact mirror image of buyer (zero-sum)	

⚠ TRAP

Exercise decision ignores premium (sunk) — exercise whenever ITM, even if net result is still a loss.

Premium & brokerage are per share × **lot**. European = exercise only on expiry; American = any time.

8 Strategies & break-evens

exam aid — not described in SM; PYQs test the pay-offs

Strategy	Position	Break-even / view
Long straddle	Buy call + buy put, **same K**	K ± total premium; expects big move either way
Long strangle	Buy OTM call (K_c) + OTM put (K_p), $K_c > K_p$	Upper = K_c + total premium; lower = K_p − total premium
Strip	1 call + 2 puts (same K)	Bearish volatility bet
Strap	2 calls + 1 put	Bullish volatility bet
Bull spread	Buy low-K call, sell high-K call	Limited gain / loss; moderately bullish
Bear spread	Buy high-K put, sell low-K put (or calls reversed)	Moderately bearish
Butterfly	Buy 1 low-K + 1 high-K call, sell 2 middle-K calls	Expects price to stay near middle K
Covered call	Long stock / long futures + **sell call**	Earns premium; caps upside (Jan 2026 Q3(a))
Protective put	Long stock + buy put	Floor on loss (insurance)

Long strangle: call 620 (premium 25) + put 480 (premium 15); total premium 40; BE 660 / 440				
Expiry price	700	550	400	660 / 440
Net P/L per share	+40	**−40** (max loss)	+40	0

⑨ Binomial — one step (three methods, same answer)

$S = 100$; up 120, down 80; $K = 100$; r_f 10% for the period.

Method	Working	Call
Risk-neutral	$p = (S(1+r) - S_d) \div (S_u - S_d) = (110 - 80)/40 = 0.75$; $C = (0.75 \times 20 + 0.25 \times 0) \div 1.10$	**13.64**
Delta / replicating	$\Delta = (C_u - C_d)/(S_u - S_d) = 20/40 = 0.5$; borrow $B = (\Delta S_d - C_d)/1.1 = 36.36$; $C = \Delta S - B$	**13.64**
Put (parity)	$P = C - S + K/(1 + r) = 13.64 - 100 + 90.91$	**4.55**

Two-step: $u = 1.1$, $d = 0.9$, r 5% per step → $p = 0.75$; terminal call values 21, 0, 0 → $C = 0.75^2 \times 21 \div 1.05^2$ = **10.71**. Work backwards node by node (needed for American puts: compare early exercise at each node).

⑩ Black-Scholes (European, non-dividend)

$$C = S \cdot N(d_1) - K \cdot e^{-rt} \cdot N(d_2)$$

$d_1 = [\ln(S/K) + (r + \sigma^2/2)t] \div \sigma\sqrt{t}$ · $d_2 = d_1 - \sigma\sqrt{t}$

$P = C - S + K \cdot e^{-rt}$ (put-call parity) or $P = K \cdot e^{-rt} N(-d_2) - S \cdot N(-d_1)$

$N(d_1)$ = hedge ratio / delta · **$N(d_2)$ = probability call finishes ITM** · $N(-d) = 1 - N(d)$ · interpolate N() from the table

May 2026 Q6(b): S 120, K 115, 50 days, r 8%, σ 30%, d_1 0.5375, e^{-rt} 0.9891 → $N(d_1)$ **0.7045** (hedge ratio ≈ 70%); $d_2 = 0.5375 - 0.30\sqrt{(50/365)}$ = **0.4264**; $N(d_2)$ **0.6651** (≈ 67%); $C = 84.54 - 75.65$ = **₹8.89**; $P = -120 + 8.89 + 113.75$ = **₹2.63** (ICAI). Verified by code: C 8.89, P 2.64 (rounding).

⑪ BSM assumptions, factors & Greeks (SM)

Assumptions (7): European options · no transaction costs · short-term interest rates known & constant · stocks pay no dividend · stock price follows a random walk · returns normally distributed · variance of return constant over option life.

Factor affecting value (SM)	Effect
(a) Price of underlying	Price ↑ → calls ↑, puts ↓
(b) Time till expiry	Option may become worthless at expiry — time value decays
(c) Volatility	Statistical / historical volatility (SV); higher volatility → higher value
(d) Interest rate	Higher rates → lower PV of exercise price (call ↑, put ↓)

Greek	SM definition
Delta	Degree to which option price moves for a small change in underlying price
Gamma	How fast delta changes — "delta of the delta"; keep small for delta hedging; adjusted only with options
Theta	Change in option price for a **one-day decrease** in time to expiry — time decay
Rho	Change in option price for a **one percentage point** change in risk-free rate
Vega	Sensitivity of option value to change in volatility; used for hedging

REPEAT R13

12 Real options — types (SM)

Option	SM meaning	Like
Growth	Strategic investments: define competitive position, gain knowledge about project profitability, give manufacturing flexibility	European call
Abandonment	Exercised if value from project's assets > PV of continuing the project	American put
Timing	Third choice besides accept / reject: **delay** the decision (when to invest)	American call

Valuation methods = same as financial options (binomial, risk-neutral, BSM). **Expanded NPV = static NPV + option value.**

13 Real vs financial options (SM table — Jan 2026, 4 marks)

Basis	Financial options	Real options
Underlying	Assets normally traded — shares, bonds, commodities	Projects — not traded
Pay-off	Specified in contract → fixed	Estimated from project cash flows → can vary
Exercise period	Short — maximum up to 1 year	Starts from end of 1st year; longer
Approach	Traded → **"Priced"**	Used for decisions → **"Valued"**

14 Risk-neutral real-option numbers (app-computed)

Case	Working
Abandonment: PV now 100; next year 140 or 70; salvage 90; r_f 10%	$p = (110 - 70)/(140 - 70) = 0.5714$; value = 0.4286 × 20 ÷ 1.1 = **₹7.79**
Option to wait: invest 100; PV now 105; next year 130 or 85; r_f 8%	$p = 0.6311$; value = 0.6311 × 30 ÷ 1.08 = **₹17.53** > NPV 5 → **wait**

15 Exotic options (SM 7.5 — 9 types)

Differ from plain vanilla in pay-off & time of exercise; more complex; mostly **OTC**.

Chooser	After a specified period buyer decides whether it is a call or a put
Compound	"Split-fee" / option on an option
Barrier	Activated only if underlying reaches a certain price in a set period
Asian	Pay-off on **average** price over a period
Look-back	Holder chooses most favourable strike from min / max price during life
Basket	Value depends on a portfolio (basket)
Bermuda	Compromise between European & American
Binary / digital	Pay-off on happening of a specific event
Spread	Pay-off on difference between prices of two underlyings

⑥ Index futures (SM 5.2)

- → Based on level of an index (Nifty 50, Sensex); used to **speculate** on the market or **hedge** a portfolio against decline
- → Not based on tangible goods → **all settlements in cash**; traded in number of contracts; value = index × multiplier (SM: 1 contract = 50 units × Nifty)
- → Margin is **continuous** — computed every business day; position squared off at closing price on expiry (SM: last Thursday of the month)
- → Purposes of futures trading (SM): **speculation** and **hedging**. Index futures hedge only **systematic** risk
- → SM TYK theory: reasons index futures became more popular than stock futures; short note on marking to market

⑯ Credit derivatives (SM 8)

- → **CDOs** — backed by pool of bonds, ABS, REITs, other CDOs (securitisation is backed by loans / receivables); covers CBOs & CLOs. **Types**: **Cash-flow CDO** (backed by cash-market debt; relies on collateral's cash to pay SPV securities) · **Synthetic CDO** (credit risk transferred **without** transferring assets; unfunded (CDS only) / fully funded (CLN) / partially funded) · **Arbitrage CDO** (issuer captures spread between collateral return and borrowing cost + management fee)
- → **Risks in CDOs**: default (hits equity / junior tranche first), interest-rate (basis), liquidity, prepayment, reinvestment, foreign-exchange
- → **CDS** = Credit (loan) + Default (non-payment) + Swap (exchange of risk): buyer pays periodic premium; seller pays on default. **Features**: non-standardised private (forward-type) contract; not exchange-traded; ISDA guidelines; can be bought from a third party; bondholder can protect against issuer insolvency; premium rises with risk. **Uses**: hedging, arbitrage, speculation. **Parties**: reference entity, buyer (investor), seller (writer). **Settlement**: physical (deliver bonds, get par) or cash (par − market value)

⑰ Commodity derivatives (SM 10)

- → **Necessary conditions (SM)**: commodity **durable & storable**; units **homogeneous**; **frequent price fluctuations with wide amplitude**; supply & demand **large**; supply flows naturally to market and there are **breakdowns in existing pattern of forward contracting**
- → **Investing**: complements investment in commodity-using companies; invest in a country's consumption & production; no dividends, only price returns. Market advantages: managers prefer derivatives to tangibles; less hassle (delivery); indirect real-asset exposure — inflation hedge
- → **Futures**: easiest & cheapest way; 3 categories — agricultural (soft), energy, metals. **Swaps**: **fixed-floating** and **commodity-for-interest**; valued as a strip of forwards; factors — hedging cost, institutional structure, liquidity, seasonality, futures bid / offer spread, brokerage, credit risk / capital / admin costs

18 Weather & electricity derivatives (SM 11–12)

→ **Weather**: underlying = weather measure (rainfall, temperature, humidity, wind); hedges **volume risk**, not price risk; first deal 1997 (Aquila Energy, OTC). **vs insurance**: insurance covers extreme low-probability events; weather derivatives cover normal-condition uncertainty too. Pricing issues: data reliability, weather forecasting, temperature modelling

→ **Electricity**: **forwards** (fixed quantity at forward price at maturity) · **futures** (standardised trading, settlement) · **swaps** (pay fixed vs floating price; = strip of forwards; locational basis swaps)

19 Derivative mishaps & 6 lessons (SM 13)

Orange County (1994)	Treasurer Robert Citron, no trading background; yield-curve play + ~$2.8 bn inverse floaters; leveraged ~$7.5 bn fund to ~$20 bn via repo; Fed raised rates 6 times in 1994 → loss **~$2 bn**; Citron jailed 1 yr, fined $1,00,000
Barings (1995)	Nick Leeson: arbitrage Nikkei 225 futures Singapore vs Osaka turned speculative long; also ran **back office**, hid losses in error account **'8888'**; loss $827 mn (SM also cites $1.4 bn); 233-year-old bank bankrupt; ING bought it for **£1**. Led to operational-risk capital in Basel II
P&G & Gibson Greetings (1994)	Bankers Trust sold complex **leveraged (ratio / power) swaps**; P&G lost $106 mn; Gibson $3 mn then $16.7 mn; out-of-court settlements; BT taken over by Deutsche Bank (1999)

Lessons: (a) don't buy a derivative you don't understand (value in-house) · (b) due diligence before making Treasury a profit centre (don't link treasurer's pay to profit) · (c) specify risk limits · (d) separate **front, middle & back** offices · (e) a hedger shouldn't become a speculator · (f) stress tests, scenario analysis (VaR + scenarios, 10–20 yrs data for black swans).

20 Correlation — how Ch 9 actually connects

Ch 9 idea	The actual link (app-written cross-reference; the Ch 9 side is SM)
Index-futures beta hedge	**Ch 6**: $\beta_p = \Sigma\, w_i\beta_i$ (weighted average of stock betas) goes straight into contracts = Value × $(\beta_2 - \beta_1)$ ÷ (futures price × lot). Hedge removes only **systematic** risk — unsystematic was already diversified away (Ch 6).
Cost of carry $F = S{\cdot}e^{rt}$	**Ch 10**: interest rate parity $F = S \times (1 + r_h) \div (1 + r_f)$ is the same idea — forward = spot + carry, else arbitrage. Actual F > fair F → buy spot, sell forward/futures (cash-and-carry), in both chapters.
Put-call parity: C + PV(K) = P + S	Used in ICAI answers to get P from C (May 2026: P = −S + C + PV(K)). **Ch 5**: warrant value (MP − E) × n is the **intrinsic value of a call**; a convertible = straight bond + call on shares.
Call / put payoffs	**Ch 10**: importer (payable) buys a **call** on the currency, exporter (receivable) buys a **put** — same max(S − K, 0) / max(K − S, 0) payoff, net of premium. **Ch 12**: interest-rate cap = series of calls on the rate (borrower), floor = series of puts (lender), collar = buy cap + sell floor.
Real options	**Ch 3**: plain NPV ignores managerial flexibility; option to expand / abandon / wait adds value → project worth NPV + option value. A negative-NPV project can still be accepted.
CDO tranches	**Ch 7**: same tranching as securitisation — senior, mezzanine, equity; losses hit the equity tranche first.
Mishaps & 6 lessons	**Ch 2**: risk limits, front/middle/back-office separation, VaR + stress tests = the risk-management framework. **FR (Ind AS 109)**: derivatives are at FVTPL unless designated in hedge accounting.